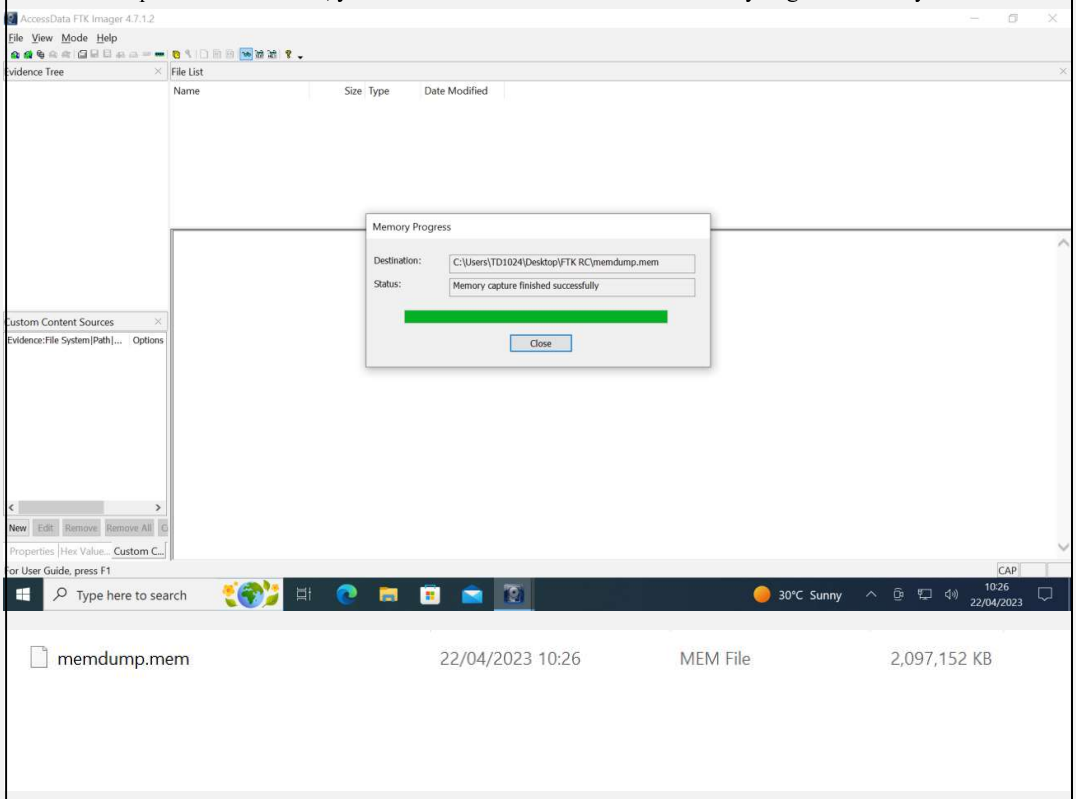
6. Once the process is finished, you can close the window and start analysing the memory file.



PRACTICAL 6

Analyzing the RAM dump using Volatility Framework to extract evidence

Volatility

Volatility is a tool for analyzing the runtime state of a system using the data found in volatile storage (RAM). It also provided a cross-platform, modular, and extensible platform to encourage further work into this exciting area of research.

Volatility is a command-based tool which is used to analyze the memory dump.

Volatility tool can be downloaded from their official website.

https://www.volatilityfoundation.org/

You can download the standalone version for Windows, Mac OS X and Linux. For standalone version there is no need to download any dependency files.

Requirements
==========
- Python 2.6 or later, but not 3.0. http://www.python.org

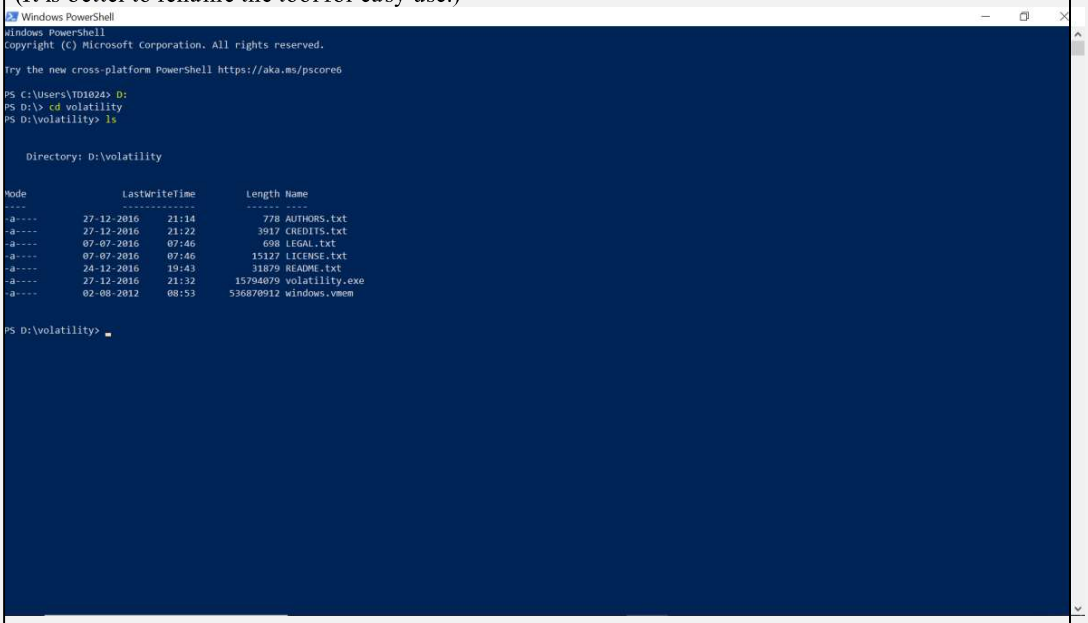
Some plugins may have other requirements which can be found at:
https://github.com/volatilityfoundation/volatility/wiki/Installation

In Windows System

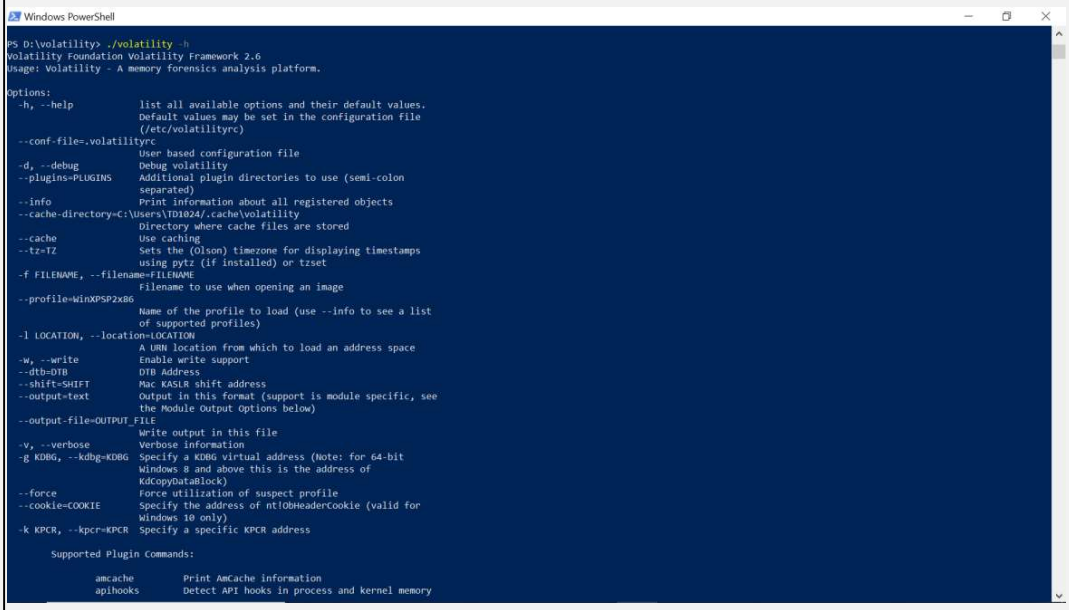
Procedure

1. Open **Windows Powershell**
2. Go to the directory where the Volatility framework is downloaded

(It is better to rename the tool for easy use.)



3. Inside powershell, type *./volatility -h* for help or *./volatility --info* for information

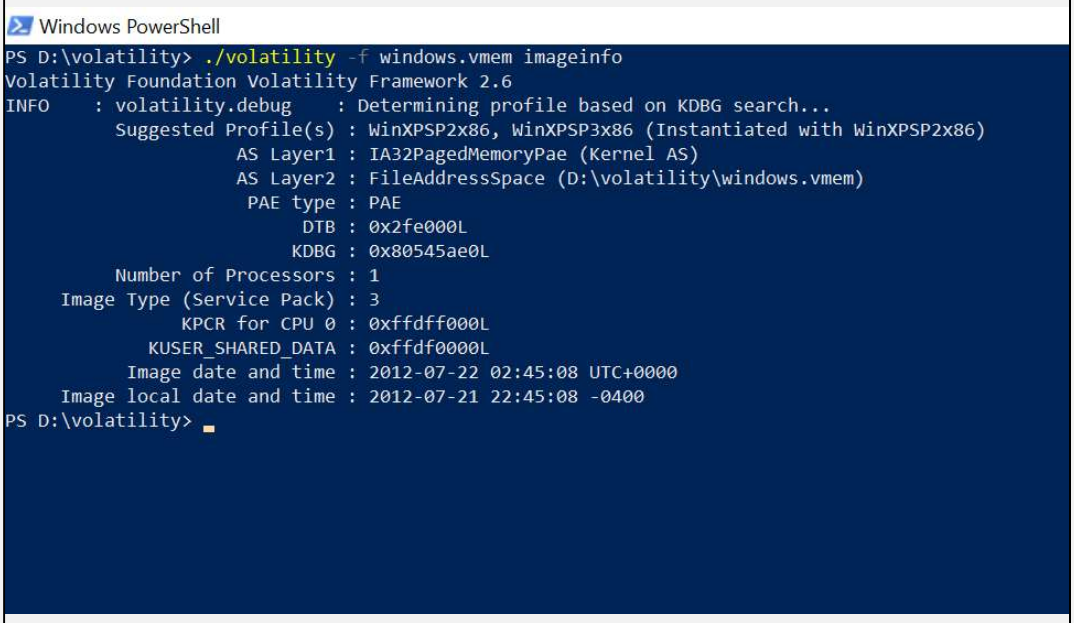


4. To analyze a memory file, Obtain the memory file and paste it in the volatility folder (The folder you have extracted after downloading the Tool).

5. Once you done that, open PowerShell and type in the command *./volatility -f windows.vmem imageinfo*

-f : Filename to use when opening an image
Imageinfo : Identify information for the image

Windows.vmem is the memory dump we are using



Here we will get the information from the memory dump. The profile, Number of processes …etc

6. We now have the computer OS from which this memory dump comes from (WinXPSP2x86). The investigation can now begin, we can specify to volatility the OS profile (--profile=WinXPSP2x86) and try to find what happened on the victim's computer.

7. To see what were the running processes use the *pslist* plugin.

./volatility -f windows.vmem --profile=WinXPSP2x86 pslist

Windows PowerShell

```
PS D:\volatility> ./volatility -f windows.vmem --profile=WinXPSP2x86 pslist
Volatility Foundation Volatility Framework 2.6
Offset(V)  Name                    PID   PPID   Thds     Hnds   Sess  Wow64 Start                          Exit
---------- -------------------- ------ ------ ------ -------- ------ ------ ------------------------------ ------------------------------
0x823c89c8 System                    4      0     53      240 ------      0
0x822f1020 smss.exe                368      4      3       19 ------      0 2012-07-22 02:42:31 UTC+0000
0x822a0598 csrss.exe               584    368      9      326      0      0 2012-07-22 02:42:32 UTC+0000
0x82298700 winlogon.exe            608    368     23      519      0      0 2012-07-22 02:42:32 UTC+0000
0x81e2ab28 services.exe            652    608     16      243      0      0 2012-07-22 02:42:32 UTC+0000
0x81e2a3b8 lsass.exe               664    608     24      330      0      0 2012-07-22 02:42:32 UTC+0000
0x82311360 svchost.exe             824    652     20      194      0      0 2012-07-22 02:42:33 UTC+0000
0x81e29ab8 svchost.exe             908    652      9      226      0      0 2012-07-22 02:42:33 UTC+0000
0x823001d0 svchost.exe            1004    652     64     1118      0      0 2012-07-22 02:42:33 UTC+0000
0x821dfda0 svchost.exe            1056    652      5       60      0      0 2012-07-22 02:42:33 UTC+0000
0x82295650 svchost.exe            1220    652     15      197      0      0 2012-07-22 02:42:35 UTC+0000
0x821dea70 explorer.exe           1484   1464     17      415      0      0 2012-07-22 02:42:36 UTC+0000
0x81eb17b8 spoolsv.exe            1512    652     14      113      0      0 2012-07-22 02:42:36 UTC+0000
0x81e7bda0 reader_sl.exe          1640   1484      5       39      0      0 2012-07-22 02:42:36 UTC+0000
0x820e8da0 alg.exe                 788    652      7      104      0      0 2012-07-22 02:43:01 UTC+0000
0x821fcda0 wuauclt.exe            1136   1004      8      173      0      0 2012-07-22 02:43:46 UTC+0000
0x8205bda0 wuauclt.exe            1588   1004      5      132      0      0 2012-07-22 02:44:01 UTC+0000
PS D:\volatility>
```

8. An alternative to the *pslist* plugin can be used to display the processes and their parent processes: *pstree*

Windows PowerShell

```
PS D:\volatility> ./volatility -f windows.vmem --profile=WinXPSP2x86 pstree
Volatility Foundation Volatility Framework 2.6
Name                                                  Pid   PPid   Thds   Hnds Time
-------------------------------------------------- ------ ------ ------ ------ ----
 0x823c89c8:System                                      4      0     53    240 1970-01-01 00:00:00 UTC+0000
. 0x822f1020:smss.exe                                 368      4      3     19 2012-07-22 02:42:31 UTC+0000
.. 0x82298700:winlogon.exe                            608    368     23    519 2012-07-22 02:42:32 UTC+0000
... 0x81e2ab28:services.exe                           652    608     16    243 2012-07-22 02:42:32 UTC+0000
.... 0x821dfda0:svchost.exe                          1056    652      5     60 2012-07-22 02:42:33 UTC+0000
.... 0x81eb17b8:spoolsv.exe                          1512    652     14    113 2012-07-22 02:42:36 UTC+0000
.... 0x81e29ab8:svchost.exe                           908    652      9    226 2012-07-22 02:42:33 UTC+0000
.... 0x823001d0:svchost.exe                          1004    652     64   1118 2012-07-22 02:42:33 UTC+0000
..... 0x8205bda0:wuauclt.exe                         1588   1004      5    132 2012-07-22 02:44:01 UTC+0000
..... 0x821fcda0:wuauclt.exe                         1136   1004      8    173 2012-07-22 02:43:46 UTC+0000
.... 0x82311360:svchost.exe                           824    652     20    194 2012-07-22 02:42:33 UTC+0000
.... 0x820e8da0:alg.exe                               788    652      7    104 2012-07-22 02:43:01 UTC+0000
.... 0x82295650:svchost.exe                          1220    652     15    197 2012-07-22 02:42:35 UTC+0000
... 0x81e2a3b8:lsass.exe                              664    608     24    330 2012-07-22 02:42:32 UTC+0000
.. 0x822a0598:csrss.exe                               584    368      9    326 2012-07-22 02:42:32 UTC+0000
 0x821dea70:explorer.exe                              1484   1464     17    415 2012-07-22 02:42:36 UTC+0000
. 0x81e7bda0:reader_sl.exe                            1640   1484      5     39 2012-07-22 02:42:36 UTC+0000
PS D:\volatility>
```

9. *psxview* will list processes that are trying to hide themselves while running on the computer, this plugin can be really useful.

Windows PowerShell

```
PS D:\volatility> ./volatility -f windows.vmem --profile=WinXPSP2x86 psxview
Volatility Foundation Volatility Framework 2.6
Offset(P)  Name                    PID pslist psscan thrdproc pspcid csrss session deskthrd ExitTime
---------- -------------------- ------ ------ ------ -------- ------ ----- ------- -------- --------
0x02498700 winlogon.exe            608 True   True   True     True   True  True    True
0x02511360 svchost.exe             824 True   True   True     True   True  True    True
0x022e8da0 alg.exe                 788 True   True   True     True   True  True    True
0x020b17b8 spoolsv.exe            1512 True   True   True     True   True  True    True
0x0202ab28 services.exe            652 True   True   True     True   True  True    True
0x02495650 svchost.exe            1220 True   True   True     True   True  True    True
0x0207bda0 reader_sl.exe          1640 True   True   True     True   True  True    True
0x025001d0 svchost.exe            1004 True   True   True     True   True  True    True
0x02029ab8 svchost.exe             908 True   True   True     True   True  True    True
0x023fcda0 wuauclt.exe            1136 True   True   True     True   True  True    True
0x0225bda0 wuauclt.exe            1588 True   True   True     True   True  True    True
```

There are no processes seem to be hidden, if so you'll see "False" in the first two columns (pslist and psscan).

10. After seeing the running processes, we can also check the running sockets and open connections on the computer. To do this we'll use these different plugins: *connscan, netscan* and *sockets*

Windows PowerShell

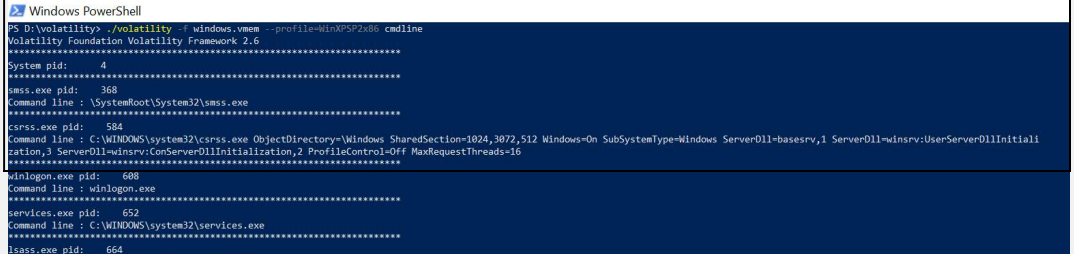
```
PS D:\volatility> ./volatility -f windows.vmem --profile=WinXPSP2x86 connscan
Volatility Foundation Volatility Framework 2.6
Offset(P)  Local Address             Remote Address            Pid
---------- ------------------------ ------------------------- ---
0x02087620 172.16.112.128:1038      41.168.5.140:8080         1484
0x023a8008 172.16.112.128:1037      125.19.103.198:8080       1484
PS D:\volatility>
```

Windows PowerShell

```
PS D:\volatility> ./volatility -f windows.vmem --profile=WinXPSP2x86 sockets
Volatility Foundation Volatility Framework 2.6
Offset(V)       PID   Port  Proto Protocol        Address         Create Time
---------- -------- ------ ------ --------------- --------------- -----------
0x81ddb780      664    500     17 UDP             0.0.0.0         2012-07-22 02:42:53 UTC+0000
0x82240d08     1484   1038      6 TCP             0.0.0.0         2012-07-22 02:44:45 UTC+0000
0x81dd7618     1220   1900     17 UDP             172.16.112.128  2012-07-22 02:43:01 UTC+0000
0x82125610      788   1028      6 TCP             127.0.0.1       2012-07-22 02:43:01 UTC+0000
0x8219cc08        4    445      6 TCP             0.0.0.0         2012-07-22 02:42:31 UTC+0000
0x81ec23b0      908    135      6 TCP             0.0.0.0         2012-07-22 02:42:33 UTC+0000
0x82276878        4    139      6 TCP             172.16.112.128  2012-07-22 02:42:38 UTC+0000
0x82277460        4    137     17 UDP             172.16.112.128  2012-07-22 02:42:38 UTC+0000
0x81e76620     1004    123     17 UDP             127.0.0.1       2012-07-22 02:43:01 UTC+0000
0x82172808      664      0    255 Reserved        0.0.0.0         2012-07-22 02:42:53 UTC+0000
0x81e3f460        4    138     17 UDP             172.16.112.128  2012-07-22 02:42:38 UTC+0000
0x821f0630     1004    123     17 UDP             172.16.112.128  2012-07-22 02:43:01 UTC+0000
0x822cd2b0     1220   1900     17 UDP             127.0.0.1       2012-07-22 02:43:01 UTC+0000
0x82172c50      664   4500     17 UDP             0.0.0.0         2012-07-22 02:42:53 UTC+0000
0x821f0d00        4    445     17 UDP             0.0.0.0         2012-07-22 02:42:31 UTC+0000
PS D:\volatility>
```

The *connscan* plugin is a scanner for TCP connections, while *sockets* will print a list of open sockets and finally *netscan* (which cannot be used in our example due to the profile used) will scan a Vista (or later) image for connections and sockets.

11. To look at the last commands ran, use *cmdscan*, *consoles* and *cmdline* plugins.

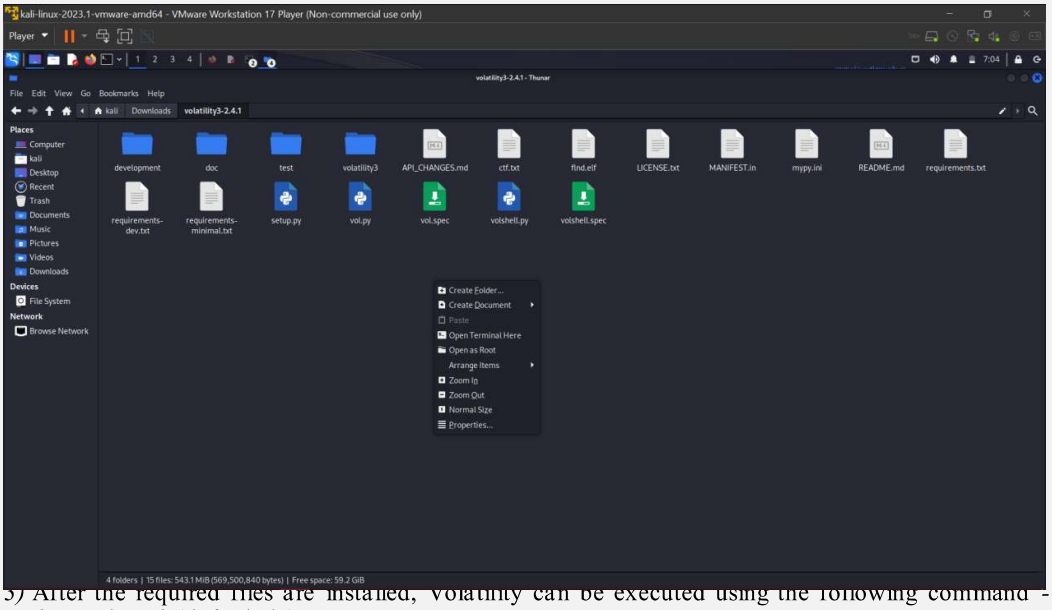


12. There are so many other plugins which can be used for analysis of memory dump. The list of commands or plugins can be seen by using the following commands

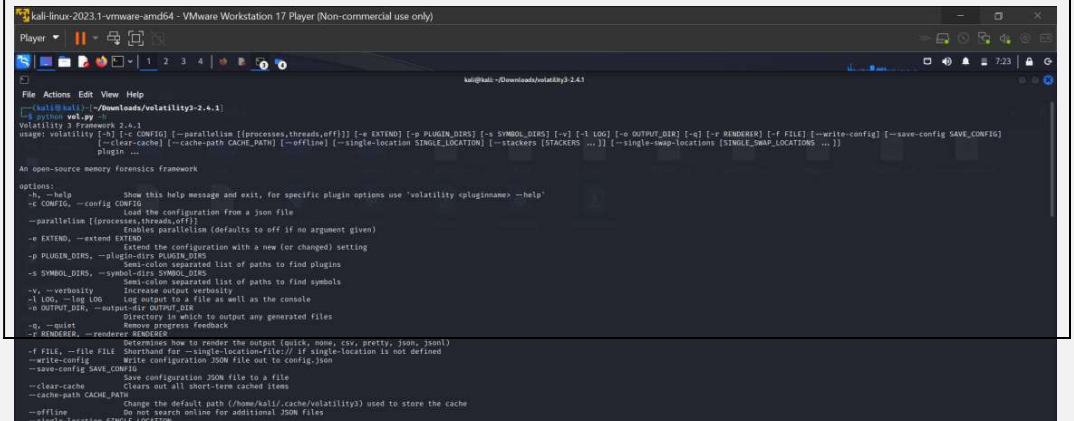
./volatility -h
./volatility --info

In Linux Systems
Download Volatility 3 v2.4.1 from the official website - https://www.volatilityfoundation.org/

Installing Volatility -
1) Right click on the extracted file and click on “open the terminal here”



3) After the required files are installed, Volatility can be executed using the following command - *python vol.py -h* (-h for help).



Rest of the process is similar to that of volatility windows. If you are analysing windows RAM dump, just add *windows."plugin"* and if you are analysing linux RAM dump, just add *linux."plugin"*

Result

RAM analysis using Volatility has successfully completed.

PRACTICAL 7

Setup SIEM tool and upload the extracted logs from windows system.

Introduction

Splunk is a popular platform for analyzing machine-generated data such as logs, events, and metrics. Splunk provides powerful search and analysis capabilities that help organizations to detect and investigate security incidents, troubleshoot operational issues, and gain insights into their IT infrastructure.
In this lab, you will learn how to set up Splunk as a Security Information and Event Management (SIEM) tool. You will learn how to install and configure Splunk, create a data source, and configure security settings to collect and analyze security events. You will also learn how to create custom alerts and dashboards to monitor and visualize security events.

Prerequisites

Before you begin this lab, you should have the following:

- A virtual or physical machine running a supported operating system (Windows, Linux, or macOS).
- Sufficient resources (CPU, RAM, and disk space) to run Splunk.
- Administrative privileges on the machine.
- Internet connectivity to download Splunk software.

Part 1: Installing and Configuring Splunk

- Download the Splunk software from the official website (https://www.splunk.com/en_us/download.html).
- Follow the instructions to install Splunk on your machine. You may need to provide administrative credentials and choose the appropriate installation options.
- After installation, launch the Splunk web interface by opening a web browser and navigating to http://localhost:8000. You should see the Splunk login page.
- Login to Splunk with the default credentials (username: admin, password: changeme).
- After login, you will be prompted to change the password. Choose a strong password and remember it for future logins.
- Once you have changed the password, you will see the Splunk Home screen. From here, you can start configuring Splunk to collect and analyze security events.

Part 2: Creating a Data Source

- To create a data source, go to the Splunk Home screen and click on "Add Data" in the "Getting Started" section.
- Select the data source that you want to collect, such as log files or network traffic.
- Follow the instructions to configure the data source. You may need to provide authentication credentials, specify the log format, and set up filters to exclude irrelevant data.
- Once you have configured the data source, Splunk will start indexing the data and making it available for search and analysis.

Part 3: Configuring Security Settings

- To configure security settings, go to the Splunk Home screen and click on "Settings" in the top menu.
- Click on "Security" to access the security settings.
- Configure the security settings according to your organization's requirements. This may include configuring authentication and authorization, enabling SSL/TLS encryption, and setting up audit logging.
- Test the security settings by logging out of Splunk and logging back in with a different user account.

Part 4: Creating Custom Alerts and Dashboards

- To create custom alerts and dashboards, go to the Splunk Home screen and click on "Create" in the top menu.
- Select "Alert" to create a new alert or "Dashboard" to create a new dashboard.
- Follow the instructions to configure the alert or dashboard. You can specify the search criteria, set up notifications, and customize the layout and visualization options.
- Once you have created the alert or dashboard, you can view it on the Splunk Home screen and monitor it for security even

Conclusion

In this lab, you learned how to set up Splunk as a Security Information and Event Management

PRACTICAL 8

Installing Wireshark and creating PCAP files for analysis. Application of Wireshark search filters.

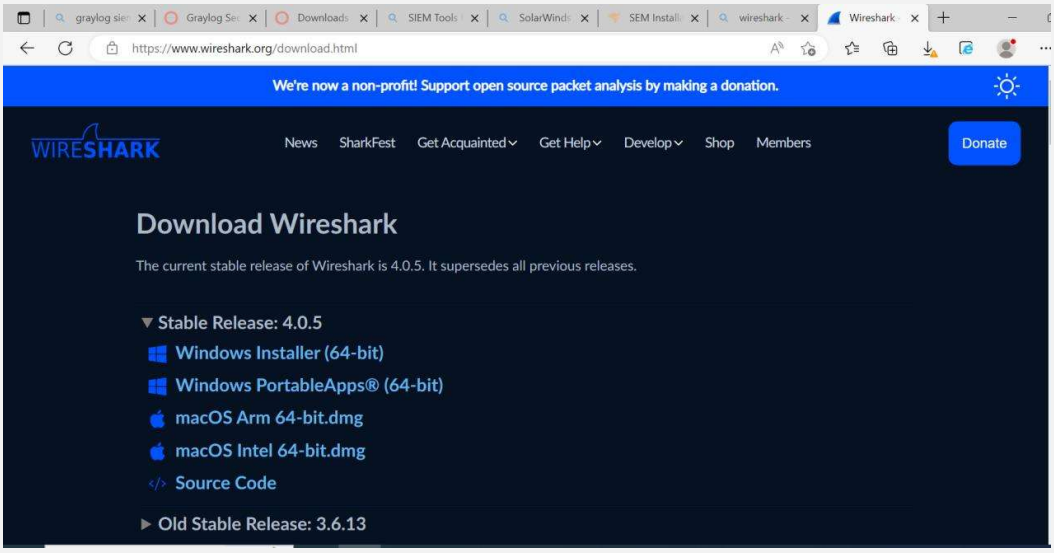
Wireshark

Wireshark is the world's foremost network protocol analyzer. It lets you see what's happening on your network at a microscopic level.

Installing Wireshark

Wireshark should be downloaded from the official website given below

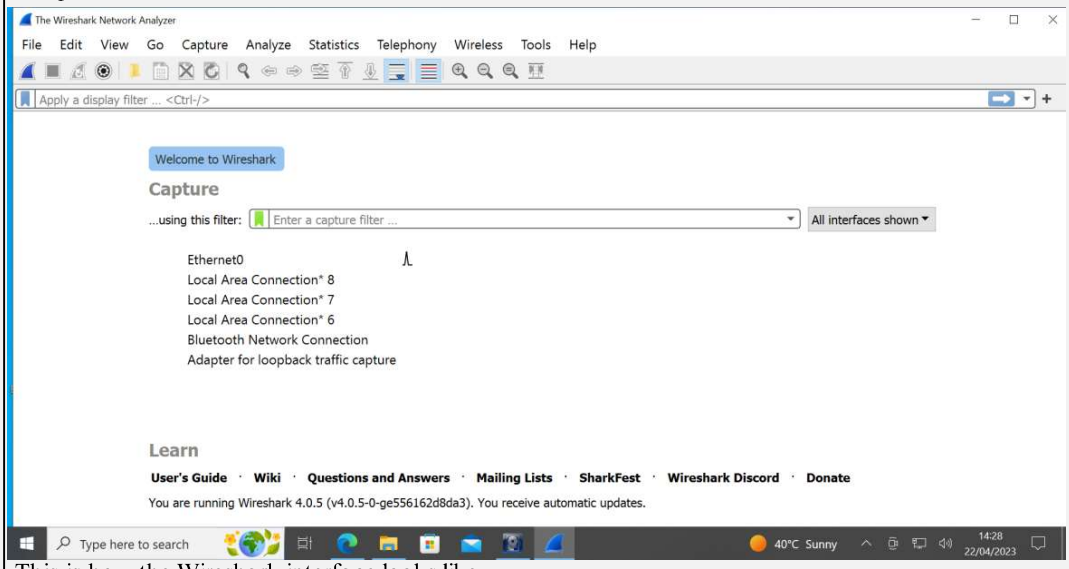
https://www.wireshark.org/download.html



Download the stable version of Wireshark from the given list.

Creating PCAP files for Analysis

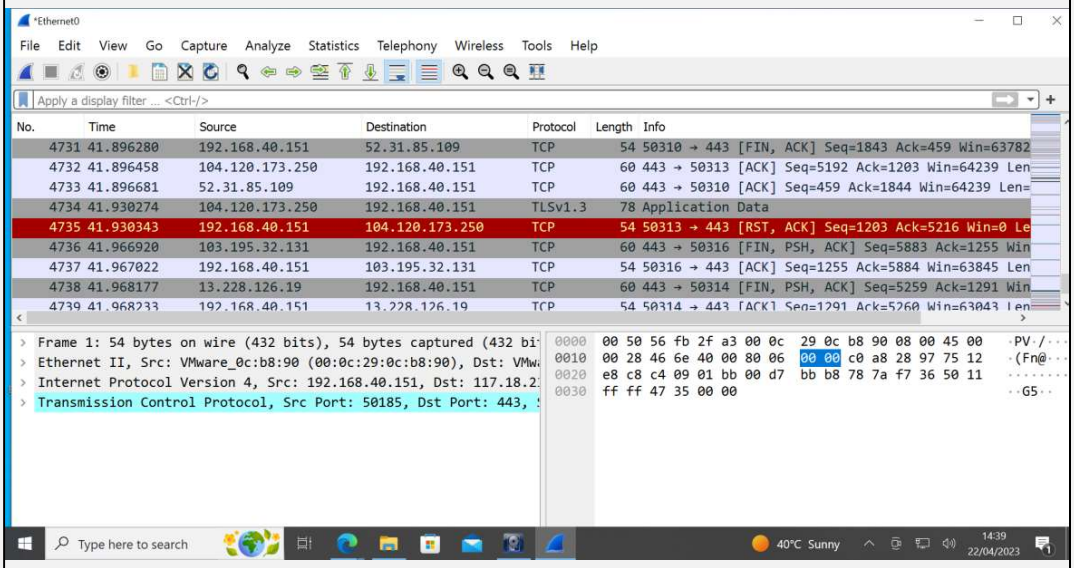
1. Open Wireshark.



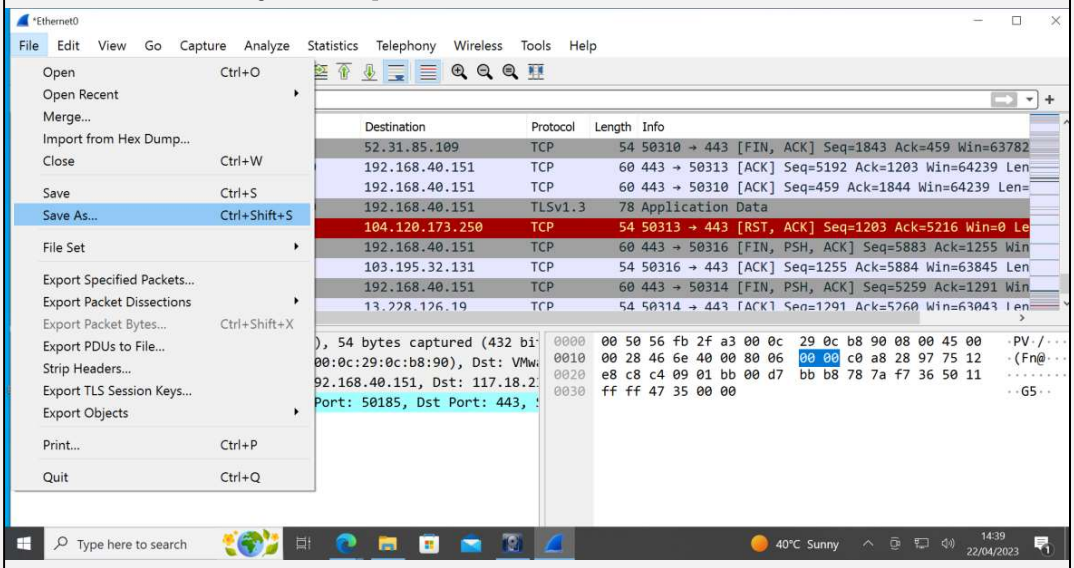
This is how the Wireshark interface looks like.

2. Select the Network you are connected to
3. Wireshark will now start to capture the packets.
4. Open Browser and search for any websites, keeping wireshark in the background.

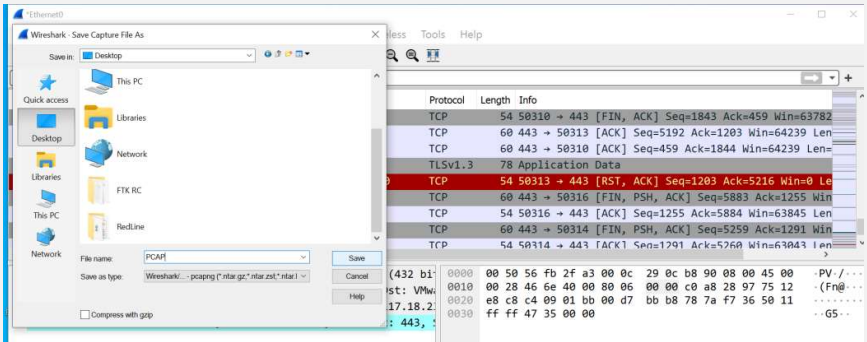
5. Wireshark will now capture the packets that is being send from the client to server and vice versa.



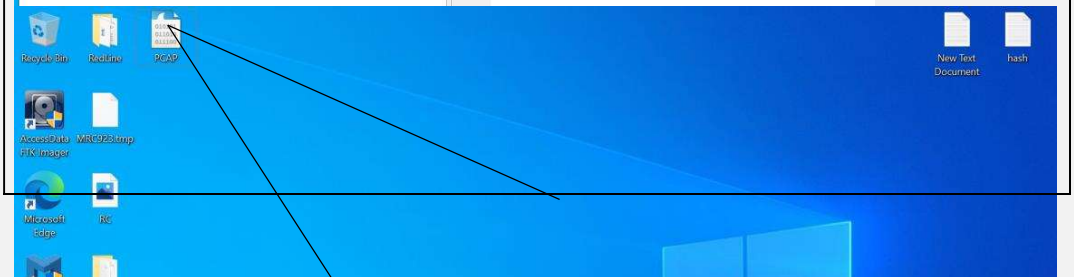
6. To create PCAP file, go to File option



7. Click on Save As….



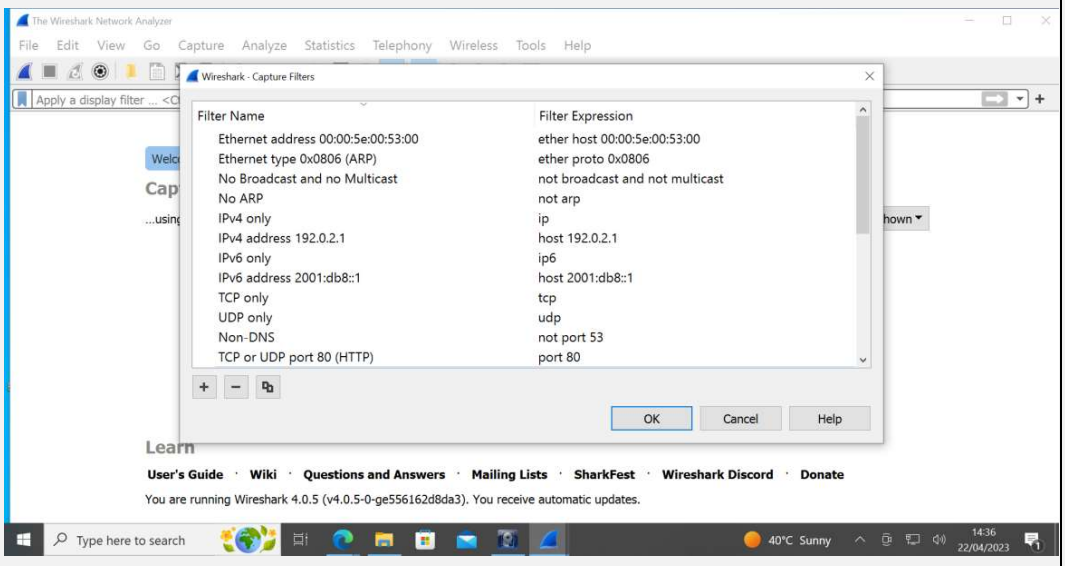
8. Type the File name and click on Save. The PCAP file will be saved on the destined Folder.



Application of Capture Filters

Capture filters limit the captured packets by the chosen filter. If the packets don't match the filter, Wireshark won't save them. Examples of capture filters include:

- host IP-address: This filter limits the captured traffic to and from the IP address
- net 192.168.0.0/24: This filter captures all traffic on the subnet
- dst host IP-address: Capture packets sent to the specified host
- port 53: Capture traffic on port 53 only
- port not 53 and not Arp: Capture all traffic except DNS and ARP traffic

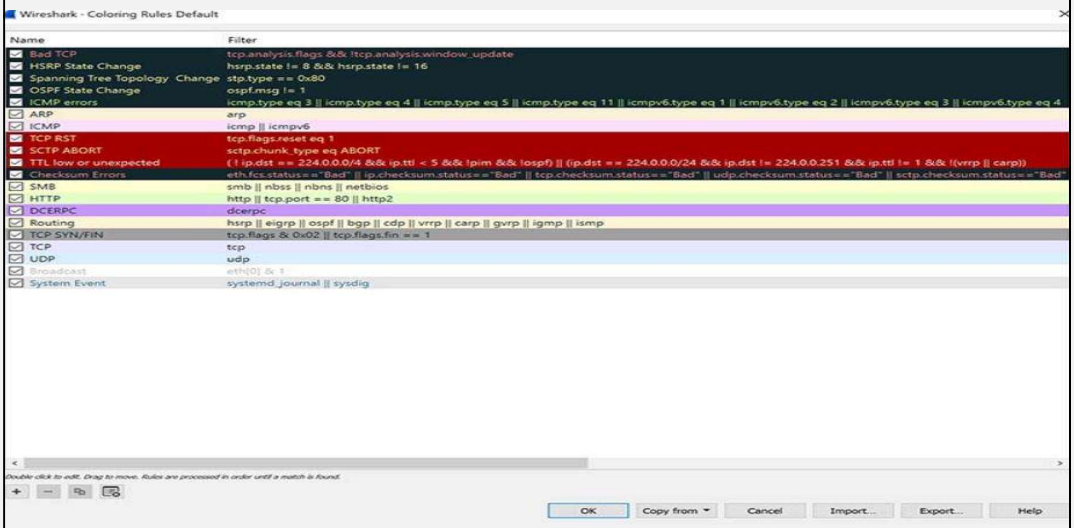


Color Coding in Wireshark

The wireshark tries to help you identify packet types by applying common-sense colour coding.

Color in Wireshark	Packet Type
Light purple	TCP
Light blue	UDP
Black	Packets with errors
Light green	HTTP traffic

You can view this by going to View >> Coloring Rules.



RESULT:-Packets were captured and analysis was studied.

PRACTICAL 9

Network malware logs analysis CTF (using Wireshark)

Introduction

Wireshark can not only be used for network analysis but also it can be used for malware analysis. It helps the investigator to determining whether the user has downloaded any malicious file from the internet or not by analyzing the network logs.